# Number systems and conversion

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## 1. Topic Overview and Learning Objectives

Number systems form the backbone of digital computing, serving as the language through which hardware interprets instructions and data. For a Class 9 student, understanding these systems is not just about memorizing conversions, but about grasping how electrical signals are translated into meaningful information. Whether it is binary, decimal, octal, or hexadecimal, each system provides a unique way to represent quantities that the computer processor can manipulate efficiently.

In the context of the Federal Board examinations, this topic is highly significant because it frequently appears in both the objective and subjective sections of the paper. Examiners often test your ability to convert between bases, which requires both accuracy and a solid grasp of the underlying mathematical logic. Mastering these conversions early in the academic year will provide you with a strong foundation for more advanced topics like digital logic and computer architecture later in your studies.

The fundamental intuition behind number systems is the concept of positional notation, where the value of a digit depends on its position relative to the radix point. By understanding the powers of the base, you can easily convert any number to decimal or vice versa. As we progress through these notes, try to visualize the computer as a machine that only understands two states, on and off, which is why binary is the primary language of the digital world.

## 2. Core Concepts and Detailed Revision Notes

### 2.1 The Decimal and Binary Number Systems

The decimal system, also known as base-10, is the system we use in our daily lives, consisting of ten digits from 0 to 9. In contrast, the binary system, or base-2, uses only two digits, 0 and 1, which correspond to the off and on states of computer transistors. Every piece of data, from images to text, is ultimately stored as a sequence of these binary digits, known as bits. When converting from binary to decimal, we use the positional weight method, where each bit is multiplied by 2 raised to the power of its position, starting from 0 on the far right. Understanding this relationship is essential because it explains how human-readable numbers are transformed into machine-readable format, allowing the computer to perform complex arithmetic operations at lightning speed.

**Key Points to Remember:**

1. Decimal system uses ten digits: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
2. Binary system uses two digits: 0 and 1.
3. The base of the decimal system is 10, while the base of binary is 2.
4. Positional value increases by powers of the base moving from right to left.
5. Binary digits are referred to as bits, and eight bits make one byte.

### 2.2 Octal and Hexadecimal Systems

To make binary numbers more manageable for humans, computer scientists use octal (base-8) and hexadecimal (base-16) systems. The octal system uses digits from 0 to 7, while the hexadecimal system uses digits 0 to 9 and letters A through F to represent values from 10 to 15. Hexadecimal is particularly important in computing because it provides a compact way to represent long binary strings, where every four bits can be represented by a single hexadecimal digit. This significantly reduces the chance of human error when writing or debugging code. Learning to convert between these bases is a core skill for any computer science student, as it bridges the gap between raw machine code and the readable formats used in memory addresses and color codes.

**Key Points to Remember:**

1. Octal system uses digits 0 through 7.
2. Hexadecimal system uses 0-9 and A-F, where A is 10 and F is 15.
3. Hexadecimal is useful for representing large binary numbers concisely.
4. Conversion between binary and hex is simple because 16 is a power of 2.
5. Each hexadecimal digit represents exactly 4 binary bits.

### 2.3 Conversion from Decimal to Other Bases

Converting a decimal number to another base requires the repeated division method. You divide the decimal number by the target base, record the remainder, and continue dividing the quotient until it reaches zero. The remainders, when read from the bottom to the top, form the equivalent number in the new base. This method is universal and works for binary, octal, and hexadecimal conversions. It is important to keep track of your remainders carefully, especially when dealing with hexadecimal, where a remainder of 10 becomes A, 11 becomes B, and so on. Practicing this method with different numbers will help you develop the speed and accuracy needed for board exams, ensuring you do not lose marks due to simple arithmetic errors during the division process.

**Key Points to Remember:**

1. Use repeated division by the target base to convert from decimal.

2. The first remainder is the least significant digit.
3. The last non-zero remainder is the most significant digit.
4. Stop dividing when the quotient becomes zero.
5. Always write the final answer in the correct base notation.

### 2.4 Conversion from Other Bases to Decimal

To convert a number from any base back to decimal, we use the positional expansion method. Each digit in the number is multiplied by the base raised to the power of its position index, starting from 0 for the rightmost digit. After calculating each product, you sum them all together to get the final decimal value. This process is very reliable and helps you verify your previous conversions. For example, if you have a binary number, you multiply each bit by the corresponding power of 2. For hexadecimal, you multiply each digit by the corresponding power of 16. This systematic approach ensures that you account for every digit correctly, reducing the likelihood of mistakes in your final calculation during the board examination.

**Key Points to Remember:**

1. Multiply each digit by the base raised to its positional power.
2. Start positional indexing from 0 at the rightmost digit.
3. Sum all the resulting products to get the decimal value.
4. Remember that A=10, B=11, C=12, D=13, E=14, F=15 in hexadecimal.
5. This method works for any base system including binary, octal, and hex.

## 3. Key Definitions, Formulas, and Rules

1. **Binary Base:** Base 2, using digits 0 and 1.
2. **Decimal Base:** Base 10, using digits 0 through 9.
3. **Octal Base:** Base 8, using digits 0 through 7.
4. **Hexadecimal Base:** Base 16, using digits 0-9 and letters A-F.
5. **Positional Value:** Value = Digit - (Base ^ Position).
6. **Bit:** The smallest unit of data, representing a 0 or 1.
7. **Byte:** A group of 8 bits.
8. **Radix:** The base number of a positional numeral system.
9. **Hexadecimal A:** Represents the decimal value 10.
10. **Hexadecimal F:** Represents the decimal value 15.

## 4. Step-by-Step Worked Examples

### Example 1: Decimal to Binary Conversion

**Problem:** Convert the decimal number 13 to binary.

**Solution Steps:**

Step 1: 13 / 2 = 6, remainder 1

Step 2: 6 / 2 = 3, remainder 0

Step 3: 3 / 2 = 1, remainder 1

Step 4: 1 / 2 = 0, remainder 1

Step 5: Read remainders from bottom to top: 1101

**Final Answer:** 1101 base 2

### Example 2: Binary to Decimal Conversion

**Problem:** Convert the binary number 1011 to decimal.

**Solution Steps:**

Step 1: 1-2^3 + 0-2^2 + 1-2^1 + 1-2^0

Step 2: 8 + 0 + 2 + 1

Step 3: Sum = 11

**Final Answer:** 11 base 10

### Example 3: Decimal to Hexadecimal Conversion

**Problem:** Convert the decimal number 25 to hexadecimal.

**Solution Steps:**

Step 1: 25 / 16 = 1, remainder 9

Step 2: 1 / 16 = 0, remainder 1

Step 3: Read remainders from bottom to top: 19

**Final Answer:** 19 base 16

### Example 4: Hexadecimal to Decimal Conversion

**Problem:** Convert the hexadecimal number 1A to decimal.

**Solution Steps:**

Step 1: 1 - 16^1 + 10 - 16^0

Step 2: 16 + 10

Step 3: Sum = 26

**Final Answer:** 26 base 10

## 5. Common Exam Mistakes to Avoid

1. Forgetting to start the positional power at 0 from the right side.
2. Confusing the remainder order when converting from decimal to other bases.
3. Mistaking the hexadecimal letter values (e.g., thinking A is 11 instead of 10).
4. Failing to include the base subscript in the final answer, which is required for clarity.
5. Performing division incorrectly during the repeated division process.
6. Mixing up the binary digits 0 and 1 during long conversions.

## 6. Practice Questions and Model Answers

### Question 1

Convert 20 decimal to binary.

**Answer 1:** 20 / 2 = 10 rem 0, 10 / 2 = 5 rem 0, 5 / 2 = 2 rem 1, 2 / 2 = 1 rem 0, 1 / 2 = 0 rem 1. Answer is 10100 base 2.

### Question 2

Convert 1110 binary to decimal.

**Answer 2:** 1-8 + 1-4 + 1-2 + 0-1 = 8 + 4 + 2 + 0 = 14 base 10.

### Question 3

Convert 15 decimal to hexadecimal.

**Answer 3:** 15 / 16 = 0 rem 15. Since 15 is F, the answer is F base 16.

### Question 4

Convert 7 octal to decimal.

**Answer 4:** 7 - 8^0 = 7 base 10.

### Question 5

Convert 10111 binary to decimal.

**Answer 5:** 1-16 + 0-8 + 1-4 + 1-2 + 1-1 = 16 + 0 + 4 + 2 + 1 = 23 base 10.